Density

📂 1.4 Density 🎯 IGCSE Physics 0625 — Extended 📝 Version 1 📅 2026-06-30

📜📜 History

The concept of density dates back to **Archimedes of Syracuse** (c. 287–212 BCE), one of the greatest mathematicians and inventors of the ancient world.

The Golden Crown Problem: King Hiero II suspected a goldsmith had replaced some gold with silver in his crown. He asked Archimedes to determine whether the crown was pure gold without destroying it.

While taking a bath, Archimedes noticed that his body displaced water — and realised he could measure the crown's volume by water displacement. He was so excited that he ran through the streets naked shouting **"Eureka!"** (Greek: "I have found it!").

By comparing the crown's density to that of pure gold, he proved the goldsmith had cheated.

Myth vs Fact: The "Eureka!" story may be embellished, but Archimedes definitely used the principle of buoyancy (now called **Archimedes' Principle**) to solve the problem. The principle states that the buoyant force on an object equals the weight of the fluid it displaces.

🌍 🌍 Nature & Applications

- 🚢 **Ships float** because their average density is less than water (they are mostly hollow — filled with air)
- 🎈 **Hot air balloons rise** because hot air is less dense than cool air
- 🧊 **Ice floats on water** because ice is less dense than liquid water — this is unusual (most solids are denser than their liquid form)
- 🌊 **Ocean currents** are driven by differences in water density (temperature and salinity)
- ⚗️ **Immiscible liquids separate** by density — oil floats on water
- 🪨 **Prospectors** use density to identify minerals and ores

⚛️ ⚛️ Theory

Definition

Density is the **mass per unit volume** of a substance. It tells you how "tightly packed" the matter is.

Where:
- (rho) = density (kg/m^3 or g/cm^3)
- = mass (kg or g)
- = volume (m^3 or cm^3)

Units

Unit System	Mass	Volume	Density
SI	kg	m^3	**kg/m^3**
Practical	g	cm^3	**g/cm^3**

Conversion:

The density of water:

Densities of Common Substances

Substance	Density (g/cm^3)	Density (kg/m^3)
Air (at sea level)	0.0012	1.2
Wood (oak)	~0.75	~750
Water	1.0	1000
Ice	0.92	920
Aluminium	2.7	2700
Iron / Steel	7.8	7800
Mercury	13.6	13600
Gold	19.3	19300

Measuring Density of a Regular Solid

1. Measure mass with a balance
2. Measure dimensions with a ruler (length, width, height)
3. Calculate volume: (or for a cylinder)
4. Calculate density:

Measuring Density of an Irregular Solid

1. Measure mass with a balance
2. Measure volume by **displacement** (submerge in water in a measuring cylinder)
3.
4. Calculate density:

Measuring Density of a Liquid

1. Measure mass of an **empty** measuring cylinder
2. Add a known volume of liquid
3. Measure mass of the **filled** measuring cylinder
4. Mass of liquid = filled mass - empty mass
5.

Floating and Sinking

- An object **floats** if its density is **less than** the fluid it's in
- An object **sinks** if its density is **greater than** the fluid it's in
- An object **hovers** (is neutrally buoyant) if its density **equals** the fluid's density

$$\rho = \frac{m}{V}$$

$$\text{Density of object} \begin{cases} < \rho_{\text{fluid}} & \Rightarrow \text{floats} \\ = \rho_{\text{fluid}} & \Rightarrow \text{neutrally buoyant} \\ > \rho_{\text{fluid}} & \Rightarrow \text{sinks} \end{cases}$$

ρ m V $1\ \text{g/cm}^3 = 1000\ \text{kg/m}^3$ $1.0\ \text{g/cm}^3 = 1000\ \text{kg/m}^3$ $V = l \times w \times h$ $\pi r^2 h$ $\rho = \frac{m}{V}$ $V = V_{\text{final}} - V_{\text{initial}}$ $\rho = \frac{m}{V}$ $\rho = \frac{m}{V}$

✍️ ✍️ Worked Examples

Example 1: Basic Density Calculation

A block of aluminium has a mass of 135 g and a volume of 50 cm³. What is its density?

Solution:

In SI units:

Example 2: Finding Mass from Density

A swimming pool has dimensions 25 m × 10 m × 2 m. What is the mass of water in the pool?

Solution:

Mass of water = kg = 500 tonnes

Example 3: Irregular Solid (Displacement)

A stone has a mass of 64 g. When placed in a measuring cylinder containing 30 mL of water, the water level rises to 54 mL. What is the density of the stone?

Solution:

Example 4: Will It Float?

A cube of material has side 5 cm and mass 150 g. Will it float in water?

Solution:

Density of the cube (1.2 g/cm³) > density of water (1.0 g/cm³), so the cube will **sink**.

$$\rho = \frac{m}{V} = \frac{135}{50} = 2.7\ \mathrm{g/cm^3}$$

$$\rho = 2.7 \times 1000 = 2700\ \mathrm{kg/m^3}$$

$$V = 25 \times 10 \times 2 = 500\ \mathrm{m^3}$$

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5.0×10^5